

# A Coupled 2D NLSWE Regional-Propagation and 3D URANS–VOF Coastal-Impact Framework for Reconstructing the 2011 Tōhoku Tsunami

Research snapshot from an ongoing framework-development programme. The work presented establishes and verifies the computational foundation for future historical validation, coastal-impact studies and tsunami-defence assessment; it does not claim a fully validated reconstruction of the 2011 event.

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## Why this matters

Tsunami-defence assessment needs local impact predictions tied to a real event-scale source. Full source-to-coast 3D simulation is impractical for routine studies, while regional depth-averaged propagation cannot resolve detailed local vertical flow, free-surface deformation or structure loading.

**Core idea:** a hybrid 2D → 3D framework connects efficient regional propagation with local URANS–VOF impact response.

## Aim and objectives

**Aim:** Establish a computational framework for reconstructing and validating real tsunami events, then reuse validated event forcing for Local3D coastal-impact and tsunami-defence studies, providing a basis for future structural and data-driven optimisation.

- Reconstruct the 2011 Tōhoku source from real event/terrain data.
- Propagate the event through a verified Regional2D NLSWE corridor.
- Establish observation-based historical-validation methodology.
- Demonstrate one-way transfer into Local3D URANS–VOF.
- Define response quantities for future defence comparison.

## Figure A: Regional context

Kamaishi is used as the coastal target for source-to-coast reconstruction of the 2011 Tōhoku event. The Regional2D corridor passes through the event reference and terminates at the selected wet nearshore interface.

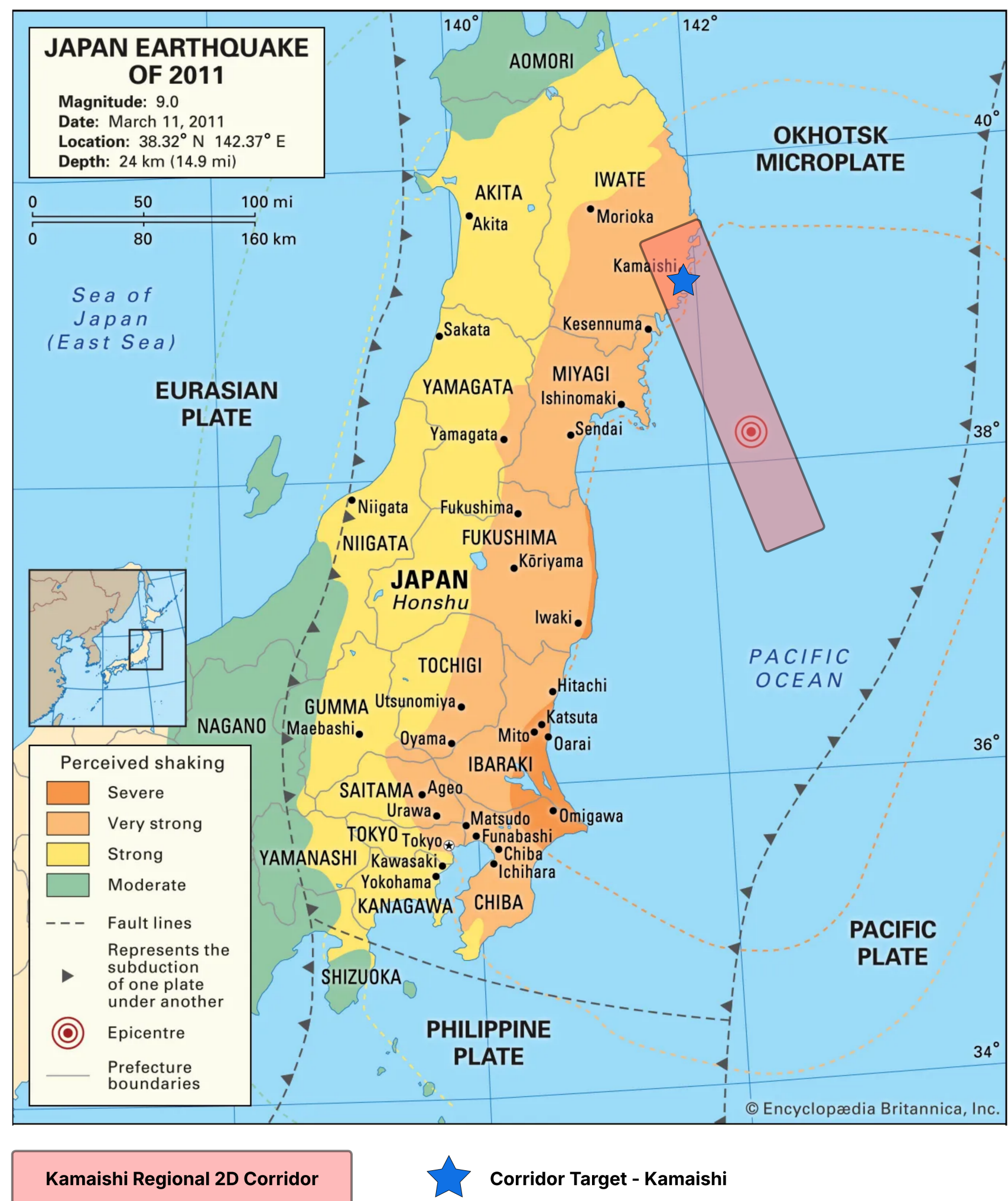


Figure A. Regional context of the 2011 Tōhoku earthquake and Kamaishi Regional2D corridor. Base map adapted from Encyclopædia Britannica; corridor and Kamaishi annotations added for this study.

## Figure B: Terrain corridor

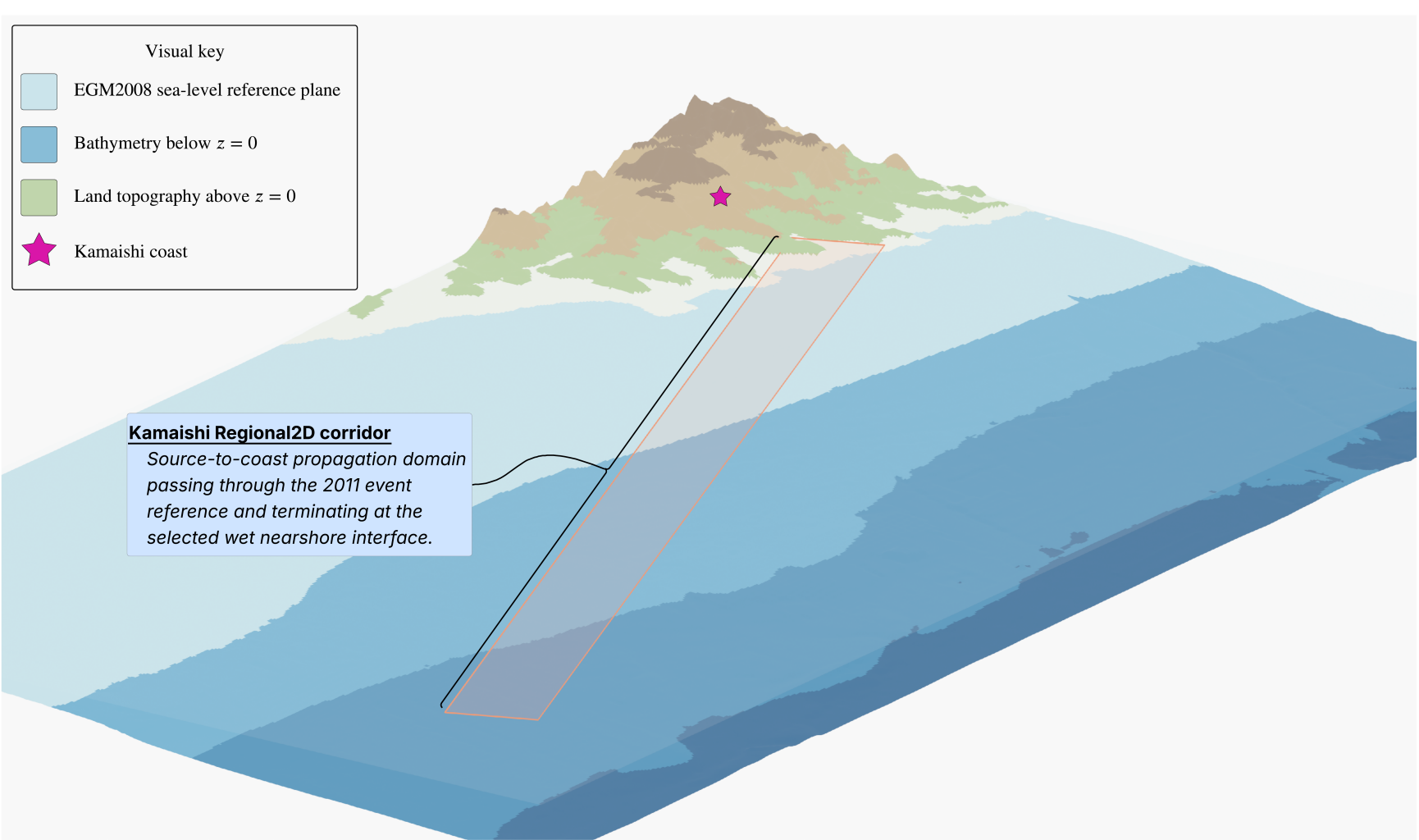
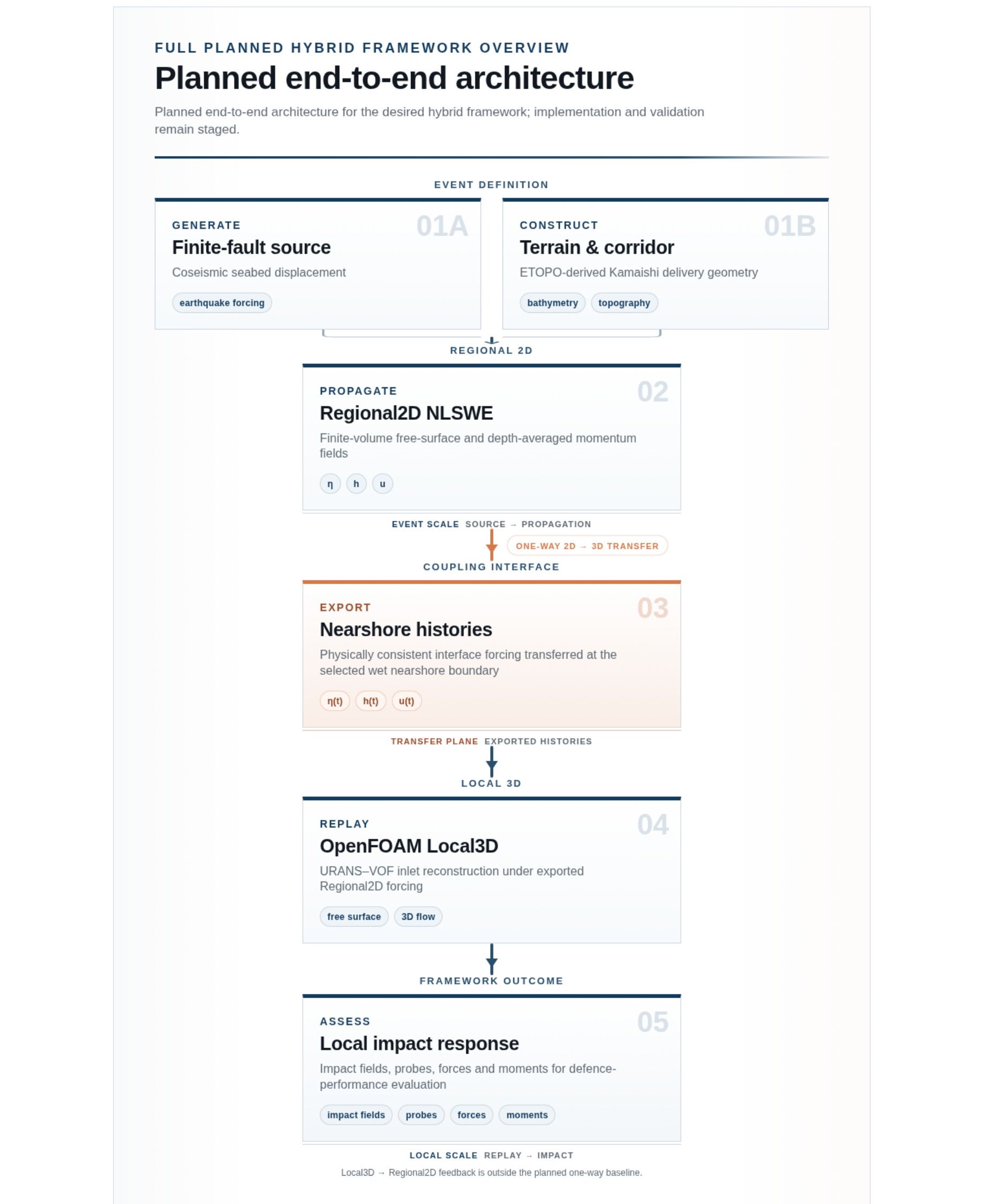


Figure B. Oblique bathymetry and topography of the Kamaishi Regional2D corridor. NOAA/NCEI ETOPO 2022, referenced to EGM2008; prepared using QGIS/GDAL and rendered in Blender 5.2 EEVEE with 4× vertical exaggeration.

## Hybrid Framework Overview



Poster-ready system architecture: event definition, Regional2D propagation, coupling export, Local3D replay and local impact response are shown as staged parts of the planned framework.

## Variables & Quantities of Interest

Solved / exported fields

### Regional propagation

- $\eta$  free-surface elevation
- $h$  water depth
- $q_x, q_y$  depth-integrated discharge
- $u, v$  depth-averaged velocity
- $b$  bed elevation
- $\Delta z_b$  source displacement

### Local flow

- $\mathbf{U}$  3D velocity
- $p$  pressure
- $\alpha_{\text{water}}$  water fraction
- $\rho, \mu, \mu_t$  density and viscosities
- $k, \omega$  SST turbulence variables
- $Re, y^+$  diagnostics

Derived / future study metrics

### Loading

- $\tau_{\text{wp}}$  wall shear stress
- $\mathbf{F}(t), \mathbf{M}(t)$  force and moment
- $J$  impulse
- pressure / viscous traction

### Coastal / defence response

- run-up; inundation; overtopping
- transmission; momentum transfer
- mechanical-energy transmission / dissipation

Response metrics define the later defence-study vocabulary; they are not all validated 2011-event results.

## Solution Algorithms

### Regional2D

repository-native solver

- event/terrain state
- wet/dry classification
- limited-linear reconstruction
- hydrostatic reconstruction
- Rusanov face flux
- bed/friction/boundary sources
- CFL timestep
- SSPRK3 update
- positivity, diagnostics, output

The repository configures and audits OpenFOAM Foundation 11; it does not reimplement OpenFOAM internal pressure/velocity, VOF or turbulence algorithms.

### Local3D

OpenFOAM Foundation 11 pathway

- regional replay data
- $\eta, q_x, q_y$  inlet reconstruction
- $\alpha_{\text{water}}/\mathbf{U}$  boundary state
- OpenFOAM incompressible URANS–VOF
- pressure/velocity solution
- VOF interface transport
- $k-\omega$  SST
- Courant/interface-Courant control
- fields, probes, force/moment output

## Verification Method

**Model consistency:** research formulation → implementation traceability → no required mismatch.

### Controlled Regional verification

- manufactured solution → FIRST-ORDER BASELINE VERIFIED
- limited-linear formulation → SECOND-ORDER LIMITED-LINEAR VERIFIED
- smooth-wave benchmark: amplitude error reduced 5.35×; phase-proxy error reduced 17.43×.

**Real-event fidelity:** event refinement and projection-only diagnosis characterise terrain/source fidelity as the dominant current limitation.

**Real-event forcing:** best available, numerically uncertain.

**Hybrid / Local evidence:** replay consistency, reflection benchmark, runtime boundedness,  $y^+$  checks, probes and force/moment extraction.

## Current Achievements

- Regional2D implementation traced consistently to the research formulation.
- First-order baseline and limited-linear second-order formulation verified.
- Real 2011 Tōhoku → Kamaishi Regional2D reconstruction completed.
- Terrain/source fidelity characterised as the present real-event spatial limitation.
- One-way Regional2D→Local3D replay framework implemented and demonstrated.
- Accepted Local3D no-defence and simple-rigid-barrier replay evidence includes free-surface/probe and force/moment outputs.
- Historical observation catalogue and validation methodology established.

## Historical Validation Framework

### Observations

- offshore waveform
- nearshore/coastal observations
- run-up
- inundation

### Metrics

- RMSE; MAE; Bias
- correlation
- peak amplitude
- peak timing

FRAMEWORK ESTABLISHED 0 DIRECT 1 PROXY 28 TARGET ONLY  
Primary validation uses no arbitrary timing or amplitude alignment. Direct collocated waveform validation for the current corridor is not complete.

## Next Steps

- Improve source/terrain fidelity and refine or extend the validation domain where needed for observation colocation.
- Perform uncalibrated historical comparison, then documented calibration and holdout validation.
- Resolve current-generation Local3D VOF replay boundedness before adopting latest Regional forcing for impact studies.
- Execute no-defence / defence impact comparisons using agreed loading, overtopping, inundation and transmission metrics.
- Extend toward FSI, structural resilience, impact-response databases and future ML optimisation/design.

## References

Encyclopædia Britannica: Japan Earthquake and Tsunami of 2011. Encyclopædia Britannica; n.d. NOAA National Centers for Environmental Information: ETOPO 2022 15 Arc-Second Global Relief Model; NOAA/NCEI. 2022. Fujii et al.; Tsunami Source of the 2011 off the Pacific Coast of Tohoku Earthquake; Earth, Planets and Space. 2011. Mori, Takahashi and The 2011 Tohoku Earthquake Tsunami Joint Survey Group: Nationwide Post Event Survey and Analysis of the 2011 Tohoku Earthquake (Tsunami); Coastal Engineering Journal; 2012. Saito et al.; Dispersion and Nonlinear Effects in the 2011 Tohoku-Oki Earthquake Tsunami; Journal of Geophysical Research: Oceans; 2014. Kurganov and Petrova; A Second-Order Well-Balanced Positivity Preserving Central-Upwind Scheme for the Saint-Venant System; Communications in Mathematical Sciences; 2007. Hirt and Nichols; Volume of Fluid (VOF) Method for the Dynamics of Free Boundaries; Journal of Computational Physics; 1981. Menter; Two-Equation Eddy-Viscosity Turbulence Models for Engineering Applications; AIAA Journal; 1994. Fujima, Masamura and Goto; Development of the 2D/3D Hybrid Model for Tsunami Numerical Simulation; Coastal Engineering Journal; 2002. Takase et al.; 2D-3D Hybrid Stabilized Finite Element Method for Tsunami Runup Simulations; Computational Mechanics; 2016. Watanabe et al.; Validation of Tsunami Numerical Simulation Models for an Idealized Coastal Industrial Site; Coastal Engineering Journal; 2022.